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DISTRIBUTION LINE VOLTAGE HARVESTER

Abstract

A distribution protection device includes an energy harvester connected in series between a first terminal and a second terminal, a switch connected in series between the first terminal and the second terminal and in parallel with the energy harvester, and switch circuitry configured to receive electrical power from the energy harvester and control operation of the switch. The energy harvester includes a first capacitive layer opposite a second capacitive layer. The first capacitive layer and the second capacitive layer form a substantially cylindrical structure that substantially envelopes the switch.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/638,190, filed Apr. 24, 2024, and U.S. Provisional Application No. 63/560,216 filed Mar. 1, 2024, the entire disclosure of both of which are incorporated by reference.

FIELD

[0002] The present disclosure relates to current interrupters and, more particularly, to self-powered current interrupters including energy harvesters for harvesting power from a power line.

SUMMARY

[0003] Distribution protection devices such as reclosers and fuses maintain the reliability and efficiency of electrical distribution systems by detecting and isolating faults. For example, distribution protection devices may detect faults such as abnormally high currents (which could indicate short circuits or overloads) and/or under-voltage, over-voltage, and/or voltage imbalance conditions (which could indicate voltage irregularities) in the electrical distribution systems. In response to detecting a fault, the distribution protection devices can interrupt current flow to the affected sections of the electrical distribution system. For example, a distribution protection device may be connected to a power line and interrupt current flow through the power line. To improve the reliability, continuity of service, and operational efficiency of the electrical distribution system, the distribution protection device may be able to automatically restore power to the affected section. For example, the distribution protection device may automatically restore power to the affected section after the fault is no longer present and/or a period of time passes.

[0004] Distribution protection devices may require electrical power to automatically restore power. For example, reclosers may include mechanical contacts that move between open and closed positions. Electrical power may be required to drive motors, hydraulic systems, and/or other mechanisms to move the contacts between the open and closed positions to connect or disconnect the power line. Distribution protection devices may also include control systems that monitor conditions of the electrical distribution system (for example, by monitoring conditions at the power line) and/or implement control logic to determine when to disconnect and when to restore power. Such control systems may also require electrical power for operation. Thus, distribution protection devices may need reliable access to electrical power for operation even when the power line is disconnected.

[0005] Some examples of distribution protection devices include energy harvesters (such as voltage or current harvesters) that harvest energy from the power line to provide electrical power for operation of the distribution protection device. However, such energy harvesters may be provided in series with the distribution protection device to ensure that the power line is disconnected when the distribution protection device is in an open position. Thus, such energy harvesters may not be able to provide electrical power to the distribution protection device when it is in the open position. To power the distribution protection device when it is in the open position, various examples of distribution protection devices include energy storage mechanisms such as supercapacitors and/or batteries. However, supercapacitors and batteries may degrade over time, potentially reducing the reliability of the distribution protection devices in some scenarios. Thus, what is needed are distribution protection devices that can harvest power from power lines when they are in the open position, eliminating the need for supercapacitors and/or batteries.

[0006] Systems, methods, apparatus, and techniques described in this specification address these and other technical problems by providing an energy harvester around the power line in parallel with the switch (such as the contacts) of the distribution protection device. In various implementations, the energy harvester may be part of or form a high-impedance circuit. Thus, even when the switch is in the open position, the high-impedance of the high-impedance circuit ensures that it does not provide an alternative path for significant current flows when the switch of the distribution protection device is open. Since only a low amount of current is able to flow through the high-impedance circuit, opening the switch effectively disconnects the power line. However, the energy harvester is able to harvest sufficient electrical energy through the high-impedance circuit to power the mechanisms and/or control systems of the distribution protection device even when the switch is open. Thus, the energy harvester is able to provide power for the distribution protection device when the switch is open without requiring supercapacitors and/or batteries.

[0007] A distribution protection device includes an energy harvester connected in series between a first terminal and a second terminal, a switch connected in series between the first terminal and the second terminal and in parallel with the energy harvester, and switch circuitry configured to receive electrical power from the energy harvester and control operation of the switch. The energy harvester includes a first capacitive layer opposite a second capacitive layer. The first capacitive layer and the second capacitive layer form a substantially cylindrical structure that substantially envelopes the switch.

[0008] In other features, the energy harvester includes a dielectric layer positioned between the first capacitive layer and the second capacitive layer. In other features, the energy harvester is part of a high-impedance circuit between the first terminal and the second terminal and the high-impedance circuit includes energy harvester circuitry configured to receive alternating current from the energy harvester and provide direct current to the switch circuitry. In other features, the energy harvester circuitry is configured to increase an impedance of the high-impedance circuit. In other features, the energy harvester includes a dielectric layer covering the first capacitive layer and the second capacitive layer. In other features, the energy harvester, the switch, and the switch circuitry are positioned within a housing. In other features, the switch is a vacuum interrupter.

[0009] A distribution protection device includes a power line extending between a first terminal and a second terminal, a switch connected in series between the first terminal and the second terminal, a capacitive core extending around the power line and connected in parallel with the switch. The switch is configured to substantially interrupt a flow of electrical power between the first terminal and the second terminal in an open position. The capacitive core is configured to continue harvesting electrical power from the power line when the switch is in the open position.

[0010] In other features, the capacitive core includes one or more dielectric layers sandwiched between one or more capacitive layers. In other features, the capacitive core includes a plurality of dielectric layers sandwiched between a plurality of capacitive layers. In other features, the plurality of capacitive layers forms a voltage divider. In other features, the capacitive core includes a first plurality of capacitive sections disposed on a first side of a dielectric layer and a second plurality of capacitive sections disposed on a second side of the dielectric layer. In other features, the first plurality of capacitive sections overlaps with the second plurality of capacitive sections. In other features, the capacitive core is part of a first circuit and the switch is part of a second circuit. The first circuit has a higher impedance than the second circuit. In other features, the capacitive core and the switch are disposed within a housing. In other features, the housing is a vacuum bottle. In other features, the first terminal and the second terminal are configured to be connected to an electrical distribution system power line.

[0011] A capacitive core for a distribution protection device includes one or more dielectric layers sandwiched between one or more capacitive layers. The one or more dielectric layers and the one or more capacitive layers are configured to envelop a switch configured to substantially interrupt a flow of electrical energy through a power line. The one or more dielectric layers and the one or more capacitive layers are configured to harvest electrical power from the power line. The

capacitive core is configured to form part of a first circuit extending between a first terminal and a second terminal of the distribution protection device. The distribution protection device includes a second circuit configured to interrupt a flow of electrical power between the first terminal and the second terminal. The first circuit is in parallel with the second circuit. The first circuit has a higher impedance than the second circuit. The one or more dielectric layers and the one or more capacitive layers are configured to continue harvesting electrical power from the power line when the second circuit interrupts the flow of electrical power between the first terminal and the second terminal.
[0012] In other features, the one or more dielectric layers and the one or more capacitive layers are configured to provide the harvested electrical power to the second circuit when the second circuit interrupts the flow of electrical power between the first terminal and the second terminal. In other features, the first circuit is configured to receive alternating current from the one or more dielectric layers and the one or more capacitive layers and provide direct current to the second circuit.
[0013] Other examples, embodiments, features, and aspects will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] FIG. **1** is an isometric view of an example distribution protection device according to some embodiments.
[0015] FIG. **2** is a front view of the distribution cutout of FIG. **1**, according to some embodiments.
[0016] FIG. **3** is a front view of the circuit interrupter of FIG. **1**, according to some embodiments.
[0017] FIG. **4** is an exploded view of the circuit interrupter of FIGS. **1** and **3**, according to some embodiments.
[0018] FIG. **5** is a front view of the internal assembly of the circuit interrupter of FIGS. **1**, **3**, and **4**, according to some embodiments.
[0019] FIG. **6** is an exploded view of the vacuum interrupter and voltage harvester assembly of FIG. **5**, according to some embodiments.
[0020] FIG. **7** is a schematic diagram showing an example of the distribution protection device connected to a power line of an electrical distribution system, according to some embodiments.
[0021] FIG. **8** is a front view of the voltage harvester assembly of FIG. **6**, according to some embodiments.
[0022] FIG. **9** is a cross-sectional view of the voltage harvester assembly of FIG. **8** taken at section line **8-8**, according to some embodiments.
[0023] FIG. **10** is a front view of the capacitive core of FIG. **9**, according to some embodiments.
[0024] FIG. **11** is a cross-sectional view of the capacitive core of FIG. **9** taken at section line **10-10**, according to some embodiments.
[0025] FIG. **12** is a schematic illustration showing details associated with some examples of the capacitive core of FIGS. **10** and **11**, according to some embodiments.
[0026] FIG. **13** is a schematic illustration showing details associated with some examples of the capacitive core of FIGS. **10** and **11**, according to some embodiments.
[0027] FIG. **14** is a schematic illustration showing details associated with some examples of the capacitive core of FIGS. **10** and **11**, according to some embodiments.
[0028] FIG. **15** is an isometric view of an example of the capacitive core of FIGS. **10** and **11**, according to some embodiments.
[0029] FIG. **16** is a cross-sectional view of the capacitive portion of the capacitive core of FIG. **15**, taken at section line **15-15**, according to some embodiments.
[0030] In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

[0031] FIG. **1** is an isometric view of an example distribution protection device **100**. In various implementations, the distribution protection device **100** may be implemented as a fuse, circuit breaker, recloser, and/or relay. For example, the distribution protection device **100** may be implemented as a self-powered recloser (such as a cutout-mounted single-phase self-powered recloser) and/or as a vacuum-bottle-based resettable fuse. In the example of FIG. **1**, the distribution protection device **100** is implemented as a cutout-mounted recloser. As shown in FIG. **1**, some implementations of the distribution protection device **100** includes a circuit interrupter **102** mounted to a distribution cutout **104**. As will be described in this specification, the circuit interrupter **102** may selectively interrupt the flow of electricity through a power line **106**.

[0032] FIG. **2** is a front view of the distribution cutout **104** of FIG. **1**. As illustrated in the example of FIG. **2**, the distribution cutout **104** may include a terminal such as an upper contact **202** for electrically connecting the circuit interrupter **102** to the power line **106**, a terminal such as a lower contact **206** for electrically connecting the circuit interrupter **102** to the power line **106**. In various implementations, the lower contact **206** also functions as a mounting bracket for mechanically coupling the circuit interrupter **102** to the distribution cutout **104**.

[0033] FIG. **3** is a front view of the circuit interrupter **102** of FIG. **1**. FIG. **4** is an exploded view of the circuit interrupter **102** of FIGS. **1** and **3**. As illustrated in the example of FIGS. **3** and **4**, the circuit interrupter **102** may include an internal assembly **402** positioned within a top housing **404** and a bottom housing **406**. The circuit interrupter **102** may also include a top cap **407** that secures a hook **408** and/or a contact connector **410** to the top housing **404**. The top cap **407** may also protect the internal assembly **402** and/or other contents of the top housing **404** and the bottom housing **406** from the external environment. In various implementations, the top cap **407** mechanically couples the contact connector **410** to the upper contact **202** and the contact connector **410** electrically connects the internal assembly **402** to the power line **106** (for example, via the upper contact **202**). In some examples, the hook **408** may be used to reposition the circuit interrupter **102** (for example, by closing the circuit interrupter **102** into the cutout **104**) and/or to assist in opening the circuit interrupter **102** for maintenance or inspection (for example, a loadbreaker tool may be attached to the hook **408** to disconnect and break the circuit between the circuit interrupter **102** and the upper contact **202**—for example, by breaking the connection between the upper contact **202** and the top cap **407**).

[0034] A trunnion **412** may be mechanically coupled to the top housing **404** and mechanically couple the top housing **404** to the lower contact **206**. In some examples, the trunnion **412** electrically connects the internal assembly **402** to the power line **106** (for example, via the lower contact **206**). In various implementations examples, a manual operating handle **414** is mechanically coupled to the bottom housing **406** and allows an operator to manually open or close the circuitry of the internal assembly **402** (e.g., to manually open or close the circuit formed by the power line **106** and the circuit interrupter **102**). In some examples, a time current characteristic (TCC) selector **416** provides a user interface that allows the operator to adjust operational settings of the circuit interrupter **102** (e.g., the timing and/or conditions under which the circuit interrupter **102** automatically recloses after detecting a fault).

[0035] FIG. **5** is a front view of the internal assembly **402** of the circuit interrupter **102** of FIGS. **1**, **3**, and **4**. As illustrated in the example of FIG. **5**, the circuit interrupter **102** may include a vacuum interrupter and voltage harvester assembly **502**, a current transformer holder **504**, a sense current transformer **506**, a current harvester **508**, a drive assembly **510**, a bistable magnetic actuator **512**, and a control unit **514**. FIG. **6** is an exploded view of the vacuum interrupter and voltage harvester assembly **502** of FIG. **5**. As shown in the example of FIG. **6**, the vacuum interrupter and voltage harvester assembly **502** may include a vacuum interrupter **602** connected in series with a power line **604**. The vacuum interrupter **602** may have a substantially cylindrical shape, and a voltage harvester assembly **606** may be positioned around the vacuum interrupter **602**. For example, the voltage harvester assembly **606** may have a substantially hollow cylinder shape and the vacuum interrupter **602** may be positioned within the inner radius (or inner diameter) of the voltage harvester assembly **606** (e.g., within the hollow central portion of the voltage harvester assembly **606**).

[0036] The vacuum interrupter **602** and the voltage harvester assembly **606** may be mechanically secured between a top sleeve **608** and a bottom sleeve **610**. In various implementations, the voltage harvester assembly **606** may be detached from the vacuum interrupter **602** when the top sleeve **608** and/or the bottom sleeve **610** are removed. Thus, in some examples, the voltage harvester assembly **606** may be referred to as a hollow-cylindrical detachable capacitive harvester.

[0037] In various implementations, the vacuum interrupter **602** includes two contacts (e.g., a fixed contact and a movable contact or two movable contacts within a vacuum-sealed envelope) that open or close the electrical circuit that the vacuum interrupter **602** completes.

[0038] In some examples, the trunnion **412**, current harvester **508**, vacuum interrupter **602**, power line **604**, and top cap **407** are electrically connected in series. Thus, when the circuit interrupter **102** is closed into the distribution cutout **104** (e.g., when the upper contact **202** is electrically connected to the top cap **407** and the lower contact **206** is electrically connected to the trunnion **412**), the vacuum interrupter **602** may function as a switch that completes the electrical circuit formed by the power line **106**, the upper contact **202**, the power line **604**, the lower contact **204**, and the power line **106**. Accordingly, when the contacts of the vacuum interrupter **602** are in the open position, the vacuum interrupter **602** opens the circuit and substantially interrupts the flow of electricity through the power line **604**, which causes the circuit interrupter **102** to substantially interrupt the flow of electricity through the power line **106**. Conversely, when the contacts of the vacuum interrupter **602** are in the closed position, the vacuum interrupter **602** completes the circuit and substantially allows the flow of electricity through the power line **604**, which causes the circuit interrupter **102** to substantially allow the flow of electricity through the power line **106**.

[0039] In various implementations, the voltage harvester assembly **606** may be electrically connected in series with the power line **604** via an input terminal **612** and electrically connected to various components to be powered via an output terminal **614**. The voltage harvester assembly **606** may harvest electrical energy from the power line **604** and power various electronic components of the circuit interrupter **102**, such as the vacuum interrupter **602**.

[0040] Returning to FIG. **5**, the current transformer holder **504** may hold or house the sense current transformers **506**. The sense current transformers **506** may measure the current flowing through the power line **604**, and the current transformer holder **504** may secure the sense current transformers **506** in optimal positions for accurate current measurement. The current harvester **508** may harvest electrical energy from the power line **604** and power various components of the circuit interrupter **102**. The drive assembly **510** may be a mechanism that physically opens and closes the contacts within the vacuum interrupter **602**. The bistable magnetic actuator **512** may be a drive mechanism for the drive assembly **510** that causes the drive assembly **510** to move the contacts of the vacuum interrupter **602** between the open and closed positions. In various implementations, the bistable magnetic actuator **512** may use permanent magnets to maintain the contacts of the vacuum interrupter **602** in the open or closed position (e.g., via the drive assembly **510**) without the need for continuous power.

[0041] The control unit **514** may receive information from the sense current transformer **506** and/or other sensors of the circuit interrupter **102** and implement control logic to open or close the contacts of the vacuum interrupter **602**. For example, the control unit **514** may control the bistable magnetic actuator **512** to open and close the contacts of the vacuum interrupter **602** by driving the drive assembly **510** according to the control logic. In various implementations, the control unit **514** receives time current characteristics curves and settings via the time characteristic curve selector **416** (e.g., input by the operator) and implements the control logic according to the time current characteristic curves. In some examples, the control unit **514** communicates with external systems for monitoring and control purposes.

[0042] FIG. **7** is a schematic diagram showing an example of the distribution protection device **100** connected to the power line **106** of the electrical distribution system. As shown in FIG. **7**, the distribution protection device **100** may be connected in series with the power line **106**. Thus, the electrical path runs sequentially from the power line **106** into the distribution protection device **100** and back out into the power line **106**. In such a configuration, the distribution protection device **100** effectively becomes part of the overall circuit of the power line **106**, and electricity flowing through the power line **106** passes through the distribution protection device **100**. Thus, substantially interrupting the flow of electricity through the distribution protection device **100** may also substantially interrupt the flow of through the power line **106**.

[0043] As previously described, the vacuum interrupter **602** of the circuit interrupter **102** may include two contacts that function as a switch **702**. The switch **702** may be electrically connected in series with the power line **106**, the upper contact **202**, the power line **604**, and the lower contact **204**. Thus, when the switch **702** is open, the vacuum interrupter **602** substantially interrupts the flow of electricity through the power line **106**. Switch control circuitry **704** may control the opening and closing of the switch **702**. In various implementations, the switch control circuitry **704** may include sensors, a microprocessor and/or an electronic controller, relays, a power supply, and/or a communications interface.

[0044] In some examples, the switch control circuitry **704** includes the sense current transformer **506** and/or the control unit **514**. The sensors may detect electrical parameters such as voltage, current, temperature, and/or frequency at the power line **604**. The microprocessor and/or electronic controller can interpret signals from the sensors and, based on programmed control logic, command the bistable magnetic actuator **512** to operate the switch **702**. The power supply receives external power (for example, from the voltage harvester assembly **606**) for powering the switch control circuitry **704**. The communications interface may include interfaces for wired or wireless communications with central control systems of the electrical distribution system, allowing for remote configuration, monitoring, manual override, etc. from the central control systems.

[0045] The distribution protection device **100** may further include the voltage harvester assembly **606** and the voltage harvester circuitry **706** electrically connected in series between the upper contact **202** and the lower contact **204** and in parallel with the switch **702**. In various implementations, the voltage harvester assembly **606** may be a capacitive energy harvester connected in series with the power line **604** (e.g., via the input terminal **612**), which harvests and/or stores electrical energy from the power line **604**. For example, in various implementations, the voltage harvester assembly **606** may include a plate capacitor, such as a multi-plate capacitor). Voltage of the AC power line **604** may change direction periodically. As the voltage of the power line **604** increases in one direction, the capacitor may charge by accumulating opposite electric charges on its plates. This charging process stores energy in the electric field between the plates. When the voltage of the power line **604** decreases (e.g., reverses direction), the capacitor may discharge, releasing stored electrical energy to the voltage harvester circuitry **706** and/or the switch control circuitry **704** (for example, to power the switch control circuitry **704**). Thus, in various implementations, the voltage harvester assembly **606** harvests electrical energy from the power line **604** and provides electrical energy to the voltage harvester circuitry **706** (e.g., via the output terminal **614**).

[0046] In various implementations, the voltage harvester circuitry **706** includes a rectifier circuit for converting the AC into a direct current (DC) (for example, high-voltage low-amperage DC), and/or a DC-DC converter to step down the voltage of the DC and/or stabilize the DC output from the voltage harvester circuitry **706**. In some examples, the voltage harvester assembly **606** may include a high-impedance capacitor. In various implementations, the voltage harvester circuitry **706** includes various components for increasing impedance in the circuit formed by the voltage harvester assembly **606** and the voltage harvester circuitry **706**. For example, the voltage harvester circuitry **706** includes one or more circuits including resistors, inductors, and/or capacitors for adding resistance, inductance, and/or capacitance to increase impedance in the high-impedance circuit. Thus, the circuit including the voltage harvester assembly **606** and/or the voltage harvester circuitry **706** may be a high-impedance circuit. Because the voltage harvester assembly **606** and/or the voltage harvester circuitry **706** form a high-impedance energy harvester circuit, the high-impedance circuit does not provide an alternative path for significant current flows between the upper contact **202** and the lower contact **204** when the switch **702** is open. However, the voltage harvester assembly **606** is still able to harvest electrical energy from the power line **604** even when the switch **702** is open.

[0047] FIG. **8** is a front view of the voltage harvester assembly **606** of FIG. **6**. FIG. **9** is a cross-sectional view of the voltage harvester assembly **606** of FIG. **8** taken at section line **8-8**. As shown in the example of FIG. **9**, the voltage harvester assembly **606** may include a capacitive core **802** formed as a substantially hollow cylindrical structure disposed around a support structure **804** also formed as a substantially hollow cylinder. In various implementations, the support structure **804** may be formed from a fiberglass material. The capacitive core **802** may be electrically connected to the power line **604** via the input terminal **612** and harvest electrical energy from the power line **604**. In various implementations, the capacitive core **802** and/or the support structure **804** may be substantially encapsulated by an covering **806**.

[0048] In some examples, the covering **806** may be formed from a dielectric material, such as silicone rubber or epoxy. In various implementations, the covering **806** may be formed as an overmold for the capacitive core capacitive core **802** and/or the support structure **804**. Providing the covering **806** may improve the performance of the capacitive core **802**. For example, the covering **806** may substantially cover the capacitive core **802**, which may increase the overall capacity of the voltage harvester assembly **606** to withstand high voltages and reduce both the effects of electrical interference from external sources and any electrical interference that the voltage harvester assembly **606** may cause. The covering **806** may also provide thermal insulation for the capacitive core **802**, keeping the capacitive core **802** at stable temperatures despite fluctuating environmental conditions. The covering **806** may also provide mechanical protection (for example, impact resistance and vibration dampening) for the capacitive core **802**, reducing the risk of mechanical damage.

[0049] FIG. **10** is a front view of the capacitive core **802** of FIG. **9**. FIG. **11** is a cross-sectional view of the capacitive core **802** of FIG. **9** taken at section line **10-10**. FIG. **12** is a schematic illustration showing details associated with some examples of the capacitive core **802** of FIGS. **10** and **11**. As shown in FIG. **12**, the capacitive core **802** may include one or more dielectric layers, such as dielectric layer **1202**, sandwiched between one or more capacitive layers, such as capacitive layer **1204** and capacitive layer **1206**. In some examples, capacitive layers **1204** and **1206** and dielectric layer **1202** are arranged concentrically around a hollow central portion **1208** of the capacitive core **802**. The capacitive layers may be connected in series with the AC power line **604** (e.g., via input terminal **612**). In various implementations, the capacitive layers may be formed of a conductive material, such as a metal foil. For example, the metal foil can include an aluminum foil, a copper foil, a silver foil, a stainless steel foil, a gold foil, and/or a nickel foil. In some examples, the dielectric layers may be formed from dielectric materials, such as ceramic dielectrics, polymer dielectrics, composite dielectrics, glass, paper and cellulose-based dielectrics, etc. Examples of suitable ceramic dielectrics include barium titanate (BaTiO.sub.3), titanium dioxide (TiO.sub.2), and/or various ferroelectric ceramics. Examples of suitable polymer dielectrics include polymer films formed from polymers such as polyvinylidene fluoride (PVDF), polytetrafluoroethylene (PTFE), polyethylene terephthalate (PET), polypropylene, etc.

[0050] In various implementations, the dielectric layer **1202** may be a dielectric flexible circuit board In various implementations, the dielectric flexible circuit board is constructed of a polymer material, such as polyimide. The dielectric flexible circuit board may serve as a substrate for the capacitive layers **1204** and **1206**. For example, the capacitive layers **1204** and **1206** may be metals deposited on a respective side of the dielectric layer **1202** using a plating technique (for example, an electroplating technique or an electroless plating technique).

[0051] As previously described, capacitive layers **1204** and **1206** may be electrically connected in series with the AC power line **604**, which may be electrically connected to the upper contact **202** and the lower contact **204**. The capacitive layers **1204** and **1206** may form a cylindrical plate capacitor, which may harvest or store energy from the power line **604** and provide the stored electrical energy to the voltage harvester circuitry **706** and/or the switch control circuitry **704** (for example, via the output terminal **614** as previously described).

[0052] FIG. **13** is a schematic illustration showing details associated with some examples of the capacitive core **802** of FIGS. **10** and **11**. In various implementations, the capacitive core **802** includes multiple dielectric layers (such as dielectric layers **1302**, **1304**, **1306**, etc.) sandwiched between pairs of capacitive layers (such as capacitive layers **1308**, **1310**, **1312**, **1314**, etc.). For example, the capacitive layers and the dielectric layers may be arranged in an alternating pattern extending concentrically outward from the hollow central portion **1208**. Although three dielectric layers **1302-1306** and four capacitive layers **1308-1314** are shown in the example of FIG. **13**, other implementations of the capacitive core **802** may include any number of dielectric layers sandwiched between any number of capacitive layers. In various implementations, the capacitive layers and the dielectric layers may be formed from any of the previously described materials.

[0053] In the example of FIG. **13**, the capacitive layers may be connected in series with the power line **604** and form a cylindrical multi-plate capacitor. In various implementations, the cylindrical multi-plate capacitor may function as a voltage divider, outputting a lower voltage to the voltage harvester circuitry **706** and/or the switch control circuitry **704** than the line voltage of the power line **604**. In various implementations, dividing the voltage has technical benefits of reducing the insulation requirement (for example, the thickness of the dielectric layers) between the capacitive layers and/or reducing the complexity of voltage limiting circuitry of the distribution protection device **100** (for example, voltage harvester circuitry **706**).

[0054] FIG. **14** is a schematic illustration showing details associated with some examples of the capacitive core **802** of FIGS. **10** and **11**. In various implementations, the capacitive core **802** includes multiple dielectric layers (such as dielectric layers **1402**, **1404**, **1406**, **1408**, etc.) and multiple capacitive layers (such as capacitive layers **1410**, **1412**, **1414**, **1416**, etc.) disposed in an alternating arrangement to form a sandwiched structure. The combined structure (including the dielectric layers and the capacitive layers) may be wound in a spiral to form the capacitive core **802** (for example, defining a substantially cylindrical shape around the hollow central portion **1208**). Although four dielectric layers **1402-1408** and four capacitive layers **1410-1416** are shown in the example of FIG. **14**, other implementations of the capacitive core **802** may include any number of dielectric layers sandwiched between any number of capacitive layers. In various implementations, the capacitive layers and the dielectric layers may be formed from any of the previously described materials.

[0055] FIG. **15** is an isometric view of an example of the capacitive core **802** of FIGS. **10** and **11**. FIG. **16** is a cross-sectional view of the capacitive portion of the capacitive core **802** of FIG. **15**, taken at section line **15-15**. In various implementations, the capacitive core **802** may include multiple overlapping capacitive sections on both sides of a dielectric layer **1502**. Each capacitive portion and the dielectric layer **1502** may include any of the materials previously described with reference to the capacitive layers and dielectric layers, respectively. In some examples, the capacitive core **802** may include a capacitive portion **1504** and a capacitive portion **1602** positioned on an inner surface of the dielectric layer **1502** (for example, the surface facing towards the hollow central portion **1208**). The capacitive core **802** may include a capacitive portion **1506** and a capacitive portion **1508** positioned on an outer surface of the dielectric layer **1502** (for example, the surface facing away from the hollow central portion **1208**).

[0056] As shown in the example of FIGS. **15** and **16**, each of the capacitive portions positioned on the outer surface of the dielectric layer **1502** overlaps with at least a portion of a capacitive portion positioned on the inner surface of the dielectric layer **1502**. For example, as shown in the example of FIGS. **15** and **16**, the capacitive portions **1506** and **1508** positioned on the outer surface of the dielectric layer **1502** overlap with different portions of the capacitive portion **1504** positioned on the inner surface of the dielectric layer **1502**. The capacitive portion **1508** positioned on the outer surface of the dielectric layer **1502** also overlaps with the entirety of the capacitive portion **1602** positioned on the inner surface of the dielectric layer **1502**.

[0057] The different overlapping portions of the capacitive portions create distinct capacitive interactions, effectively forming different capacitors. For example, the capacitive portion **1506** and the overlapping portion of the capacitive portion **1504** effectively form a first capacitor, while the portion of the capacitive portion **1508** overlapping with the capacitive portion **1504** and the portion of the capacitive portion **1508** overlapping with the capacitive portion **1602** effectively form a second capacitor and a third capacitor. Thus, the multiple overlapping capacitive portions effectively form multiple capacitors to divide the voltage harvested from the power line **604**. Although two capacitive portions **1504** and **1602** are shown on the inner surface of the dielectric layer **1502** and two capacitive portions **1506** and **1508** are shown on the outer surface of the dielectric layer **1502**, any number of capacitive portions may be positioned on the inner and outer surfaces of the dielectric layer **1502** and overlap in different ways to form any number of effective capacitors.

[0058] The foregoing description is merely illustrative in nature and does not limit the scope of the disclosure or its applications. The broad teachings of the disclosure may be implemented in many different ways. While the disclosure includes some particular examples, other modifications will become apparent upon a study of the drawings, the text of this specification, and the following claims. In the written description and the claims, one or more processes within any given method may be executed in a different order—or processes may be executed concurrently or in combination with each other—without altering the principles of this disclosure. Similarly, instructions stored in a non-transitory computer-readable medium may be executed in a different order—or concurrently—without altering the principles of this disclosure. Unless otherwise indicated, the numbering or other labeling of instructions or method steps is done for convenient reference and does not necessarily indicate a fixed sequencing or ordering.

[0059] Unless the context of their usage unambiguously indicates otherwise, the articles “a,” “an,” and “the” should not be interpreted to mean “only one.” Rather, these articles should be interpreted to mean “at least one” or “one or more.” Likewise, when the terms “the” or “said” are used to refer to a noun previously introduced by the indefinite article “a” or “an,” the terms “the” or “said” should similarly be interpreted to mean “at least one” or “one or more” unless the context of their usage unambiguously indicates otherwise.

[0060] Spatial and functional relationships between elements—such as modules—are described using terms such as (but not limited to) “connected,” “engaged,” “interfaced,” and/or “coupled.” Unless explicitly described as being “direct,” relationships between elements may be direct or include intervening elements. The phrase “at least one of A, B, and C” should be construed to indicate a logical relationship (A OR B OR C), where OR is a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.” The term “set” does not necessarily exclude the empty set. For example, the term “set” may have zero elements. The term “subset” does not necessarily require a proper subset. For example, a “subset” of set A may be coextensive with set A, or include elements of set A. Furthermore, the term “subset” does not necessarily exclude the empty set.

[0061] In the figures, the directions of arrows generally demonstrate the flow of information—such as data or instructions. The direction of an arrow does not imply that information is not being transmitted in the reverse direction. For example, when information is sent from a first element to a second element, the arrow may point from the first element to the second element. However, the second element may send requests for data to the first element, and/or acknowledgements of receipt of information to the first element. Furthermore, while the figures illustrate a number of components and/or steps, any one or more of the components and/or steps may be omitted or duplicated, as suitable for the application and setting.

[0062] The term computer-readable medium does not encompass transitory electrical or electromagnetic signals or electromagnetic signals propagating through a medium-such as on an electromagnetic carrier wave. The term “computer-readable medium” is considered tangible and non-transitory. The functional blocks, flowchart elements, and message sequence charts described above serve as software specifications that can be translated into computer programs by the routine work of a skilled technician or programmer.

[0063] It should also be understood that although certain drawings illustrate hardware and software as being located within particular devices, these depictions are for illustrative purposes only. In some embodiments, the illustrated components may be combined or divided into separate software, firmware, and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device, or they may be distributed among different computing devices-such as computing devices interconnected by one or more networks or other communications systems.

[0064] In the claims, if an apparatus or system is claimed as including an electronic processor or other element configured in a certain manner, the claim or claimed element should be interpreted as meaning one or more electronic processors (or other element as appropriate). If the electronic processor (or other element) is described as being configured to make one or more determinations or one or execute one or more steps, the claim should be interpreted to mean that any combination of the one or more electronic processors (or any combination of the one or more other elements) may be configured to execute any combination of the one or more determinations (or one or more steps).

Claims

1. A distribution protection device comprising: a switch connected in series between a first terminal and a second terminal; and an energy harvester connected in series between the first terminal and the second terminal and in parallel with the switch, the energy harvester including a first capacitive layer opposite a second capacitive layer, the first capacitive layer and the second capacitive layer forming a substantially cylindrical structure that substantially envelopes the switch; switch circuitry configured to receive electrical power from the energy harvester and control operation of the switch.

2. The distribution protection device of claim 1, wherein the energy harvester forms a hollow cylindrical structure.

3. The distribution protection device of claim 1, wherein the energy harvester has an inner diameter and the switch is positioned within the inner diameter.

4. The distribution protection device of claim 1 wherein the energy harvester is detachable from the switch.

5. The distribution protection device of claim 1, wherein the energy harvester includes a dielectric layer positioned between the first capacitive layer and the second capacitive layer.

6. The distribution protection device of claim 1, wherein: wherein the energy harvester is part of a high-impedance circuit between the first terminal and the second terminal; and the high-impedance circuit includes energy harvester circuitry configured to receive alternating current from the energy harvester and provide direct current to the switch circuitry.

7. The distribution protection device of claim 6, wherein the energy harvester circuitry is configured to increase an impedance of the high-impedance circuit.
8. The distribution protection device of claim 1, wherein the energy harvester includes a dielectric layer covering the first capacitive layer and the second capacitive layer.
9. The distribution protection device of claim 1, wherein the energy harvester, the switch, and the switch circuitry are positioned within a housing.
10. The distribution protection device of claim 9, wherein the switch is a vacuum interrupter.
11. A distribution protection device comprising: a power line extending between a first terminal and a second terminal; a switch connected in series between a first terminal and a second terminal; and a capacitive core extending around the switch, the capacitive core configured to harvest electrical power from the power line and connected in parallel with the switch; and wherein the switch is configured to substantially interrupt a flow of electrical power between the first terminal and the second terminal in an open position; and wherein the capacitive core is configured to continue harvesting electrical power from the power line when the switch is in the open position.
12. The distribution protection device of claim 11, wherein the capacitive core includes one or more dielectric layers sandwiched between one or more capacitive layers.
13. The distribution protection device of claim 11, wherein the capacitive core includes a plurality of dielectric layers sandwiched between a plurality of capacitive layers.
14. The distribution protection device of claim 13, wherein the plurality of capacitive layers forms a voltage divider.
15. The distribution protection device of claim 11, wherein the capacitive core includes a first plurality of capacitive sections disposed on a first side of a dielectric layer and a second plurality of capacitive sections disposed on a second side of the dielectric layer.
16. The distribution protection device of claim 15, wherein the first plurality of capacitive sections overlaps with the second plurality of capacitive sections.
17. The distribution protection device of claim 11, wherein the capacitive core is part of a first circuit and the switch is part of a second circuit, wherein the first circuit has a higher impedance than the second circuit.
18. The distribution protection device of claim 11, wherein the capacitive core and the switch are disposed within a housing.
19. The distribution protection device of claim 18, wherein the switch is a vacuum interrupter.
20. The distribution protection device of claim 18, wherein the first terminal and the second terminal are configured to be connected to an electrical distribution system power line.
21. A capacitive core for a distribution protection device comprising: one or more dielectric layers sandwiched between one or more capacitive layers; wherein the one or more dielectric layers and the one or more capacitive layers are configured to envelop a switch configured to substantially interrupt a flow of electrical energy through a power line a power line; wherein the one or more dielectric layers and the one or more capacitive layers are configured to harvest electrical power from the power line; wherein the capacitive core is configured to form part of a first circuit extending between a first terminal and a second terminal of the distribution protection device; wherein the distribution protection device includes a second circuit configured to interrupt a flow of electrical power between the first terminal and the second terminal; wherein the first circuit is in parallel with the second circuit; wherein the first circuit has a higher impedance than the second circuit; and wherein the one or more dielectric layers and the one or more capacitive layers are configured to continue harvesting electrical power from the power line when the second circuit interrupts the flow of electrical power between the first terminal and the second terminal.
22. The capacitive core of claim 21, wherein the one or more dielectric layers and the one or more capacitive layers are configured to provide the harvested electrical power to the second circuit when the second circuit interrupts the flow of electrical power between the first terminal and the second terminal.

23. The capacitive core of claim 21, wherein the first circuit is configured to receive alternating current from the one or more dielectric layers and the one or more capacitive layers and provide direct current to the second circuit.